🚀 FEETFLOW –Tech Stack

1. 🖥️ Frontend Stack (User Interface Layer)

- ⚡ **Framework**

  Next.js (React Framework)

  - **SEO friendly (landing page)**
  - **Fast rendering**
  - **Built-in routing**
  - **Production ready**

- 🎨 Styling

  Tailwind CSS

  - Rapid UI development
  - Clean utility-based styling
  - Perfect for modern SaaS dashboards

- 🎬 Animations

  Framer Motion

  - Smooth transitions
  - Dashboard micro-interactions
  - Hero animations

- 📊 Charts & Analytics

  Recharts

  - Lightweight
  - Easy to integrate with React
  - Perfect for KPI dashboards

- 🧩 UI Components
  - Headless UI (for modals, dropdowns)
  - Heroicons or Lucide Icons

2. ⚙️ Backend Stack (Business Logic Layer)

🧠 **Runtime**

Node.js

🚦 **Framework**

Express.js

- REST API development
- Easy middleware integration
- Lightweight & scalable

🔐 **Authentication & Security**

- JWT (Access & Refresh tokens)
- bcrypt (Password hashing)
- Role-Based Access Control (RBAC)
- Helmet.js (Security headers)
- Rate limiting middleware

3. 🗄️ Database Layer

Primary Database

PostgreSQL

- Relational data (Vehicles, Trips, Drivers)
- ACID compliance
- Complex query support (for analytics)

ORM

Prisma

- Clean schema modeling
- Type-safe queries
- Easy migrations

Caching Layer

Redis

- Cache dashboard KPIs
- Reduce DB load
- Session management

4. 📡 Real-Time Capabilities

For vehicle/driver availability updates:

Socket.io

- Real-time trip status updates
- Live dashboard refresh

5. ☁️ Hosting & DevOps

Frontend Hosting

- Vercel (best for Next.js)

Backend Hosting

- Render / Railway / AWS EC2

Database Hosting

- Supabase (Postgres managed)

or

- AWS RDS

File Storage

- AWS S3 (documents, insurance, RC uploads)

6. 📈 **Analytics & Monitoring**

- Google Analytics (Landing page)
- Sentry (Error tracking)
- LogRocket (Optional session replay)

7. 🧪 Testing Stack

Frontend:

- Jest
- React Testing Library

Backend:

- Jest
- Supertest

API Testing:

- Postman

8. 🧮 Data Model Support

Core Entities:

- Users
- Vehicles
- Drivers
- Trips
- Fuel Logs
- Maintenance Logs
- Expenses

- Rules
- Alerts

Relational integrity handled via Prisma +PostgreSQL.

9. 🏗️ Architecture Overview

Frontend (Next.js)

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API Layer (Express / NestJS)

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Business Logic Layer

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PostgreSQL + Redis

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File Storage (S3)